serviceRegistryManagement MQTT/TLS/JSON

Interface Design Description

Service ID: *"serviceRegistryManagement"*

Abstract
This document describes the MQTT protocol with TLS payload security and JSON payload encoding variant of the `serviceRegistryManagement` service of the ServiceRegistry system.

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `serviceRegistryManagement` service interface of the ServiceRegistry system over MQTT, which provides operator management of the whole ServiceRegistry content: devices, systems, service definitions, service instances and interface templates.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	MQTT	3.1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `serviceRegistryManagement` service interface

This document provides the Interface Design Description IDD to the *serviceRegistryManagement – Service Description* document [5]. The operations, their semantics and the data models are the same as in the HTTP/TLS/JSON IDD [6]; this document specifies the MQTT binding.

The MQTT interface is offered only when the system runs with `mqtt.api.enabled=true`; it is then registered with the interface template `generic_mqtts` (`generic_mqtt` without TLS), the broker address and port (`mqtt.broker.address`, `mqtt.broker.port`) and the base topic `arrowhead/serviceregistry/management/`. In the Arrowhead-test-cloud-java deployment the MQTT interfaces of the core systems are disabled.

3 Service Operations

The provider subscribes to one topic per operation, the base topic followed by the operation name:

Operation	Request topic
device-query	arrowhead/serviceregistry/management/device-query
device-create	arrowhead/serviceregistry/management/device-create
device-update	arrowhead/serviceregistry/management/device-update
device-remove	arrowhead/serviceregistry/management/device-remove
system-query	arrowhead/serviceregistry/management/system-query
system-create	arrowhead/serviceregistry/management/system-create
system-update	arrowhead/serviceregistry/management/system-update
system-remove	arrowhead/serviceregistry/management/system-remove
service-definition-query	arrowhead/serviceregistry/management/service-definition-query
service-definition-create	arrowhead/serviceregistry/management/service-definition-create
service-definition-remove	arrowhead/serviceregistry/management/service-definition-remove
service-query	arrowhead/serviceregistry/management/service-query
service-create	arrowhead/serviceregistry/management/service-create
service-update	arrowhead/serviceregistry/management/service-update
service-remove	arrowhead/serviceregistry/management/service-remove
interface-template-query	arrowhead/serviceregistry/management/interface-template-query
interface-template-create	arrowhead/serviceregistry/management/interface-template-create
interface-template-remove	arrowhead/serviceregistry/management/interface-template-remove

A consumer publishes a request envelope (MqttRequestTemplate) to the topic and receives the answer (MqttResponseTemplate) on its own `responseTopic`. The `authentication` field carries the requester's identity according to the provider's authentication policy: `IDENTITY-TOKEN//<token>` (outsourced or internal policy), `SYSTEM//<name>` (declared policy) or the requester's X.509 certificate (certificate policy). The `status` of the response uses the HTTP status code numbers (200, 201, 204, 400, 401, 403, 404, 408, 423, 500, 503); on error the payload is an ErrorMessage.

```
{
  "traceId": "a1b2c3",
  "authentication": "IDENTITY-TOKEN//<token>",
  "responseTopic": "arrowhead/response/TemperatureDisplay/a1b2c3",
  "qosRequirement": 1,
  "params": {},
  "payload": {}
}
```

Listing 1: Request envelope.

```
{
  "status": 200,
  "traceId": "a1b2c3",
  "receiver": "TemperatureDisplay",
  "payload": {}
```

```
}
```

Listing 2: Response envelope.

3.1 PUBLISH arrowhead/serviceregistry/management/device-query

Operation: device-query
Input: DeviceQueryRequest
Output: DeviceListResponse

Returns a page of devices matching the filter.

```
{
  "pagination": {
    "page": 0,
    "size": 10,
    "direction": "ASC",
    "sortField": "createdAt"
  },
  "deviceNames": [
    "Raspberry1"
  ],
  "addresses": [
    "string"
  ],
  "addressType": "IPV4",
  "metadataRequirementList": [
    {
      "unit": "celsius"
    }
  ]
}
```

Listing 3: Payload of an device-query request.

3.2 PUBLISH arrowhead/serviceregistry/management/device-create

Operation: device-create
Input: DeviceListRequest
Output: DeviceListResponse

Creates devices.

```
{
  "devices": [
    {
      "name": "Raspberry1",
      "metadata": {
        "location": "hall-1"
      },
      "addresses": [
        "string"
      ]
    }
  ]
}
```

Listing 4: Payload of an device-create request.

3.3 PUBLISH arrowhead/serviceregistry/management/device-update

Operation: device-update
Input: DeviceListRequest
Output: DeviceListResponse

Updates the metadata and addresses of devices.

```
{
  "devices": [
    {
      "name": "Raspberry1",
      "metadata": {
        "location": "hall-1"
      },
      "addresses": [
        "string"
      ]
    }
  ]
}
```

Listing 5: Payload of an device-update request.

3.4 PUBLISH arrowhead/serviceregistry/management/device-remove

Operation: device-remove
Input: Array<String>

Removes devices by name.

```
[
  "TemperatureSensor"
]
```

Listing 6: Payload of an device-remove request.

3.5 PUBLISH arrowhead/serviceregistry/management/system-query

Operation: system-query
Input: SystemQueryRequest
Output: SystemListResponse

Returns a page of systems matching the filter.
Optional entries of `params`: `verbose`.

```
{
  "pagination": {
    "page": 0,
    "size": 10,
    "direction": "ASC",
    "sortField": "createdAt"
  },
  "systemNames": [
    "TemperatureSensor"
  ],
  "addresses": [
    "string"
  ],
  "addressType": "IPV4",
  "metadataRequirementList": [
```

```
    {
      "unit": "celsius"
    }
  ],
  "versions": [
    "1.0.0"
  ],
  "deviceNames": [
    "Raspberry1"
  ]
}
```

Listing 7: Payload of an system-query request.

3.6 PUBLISH arrowhead/serviceregistry/management/system-create

Operation: system-create
Input: SystemListRequest
Output: SystemListResponse

Creates systems on behalf of others.

```
{
  "systems": [
    {
      "name": "TemperatureSensor",
      "metadata": {
        "location": "hall-1"
      },
      "version": "1.0.0",
      "addresses": [
        "string"
      ],
      "deviceName": "Raspberry1"
    }
  ]
}
```

Listing 8: Payload of an system-create request.

3.7 PUBLISH arrowhead/serviceregistry/management/system-update

Operation: system-update
Input: SystemListRequest
Output: SystemListResponse

Updates systems.

```
{
  "systems": [
    {
      "name": "TemperatureSensor",
      "metadata": {
        "location": "hall-1"
      },
      "version": "1.0.0",
      "addresses": [
        "string"
      ],
      "deviceName": "Raspberry1"
    }
  ]
}
```

Listing 9: Payload of an system-update request.

3.8 PUBLISH arrowhead/serviceregistry/management/system-remove

Operation: system-remove
Input: Array<String>

Removes systems by name, with their service instances.

```
[
  "TemperatureSensor"
]
```

Listing 10: Payload of an system-remove request.

3.9 PUBLISH arrowhead/serviceregistry/management/service-definition-query

Operation: service-definition-query
Input: Page
Output: ServiceDefinitionListResponse

Returns a page of service definitions.

```
{
  "page": 0,
  "size": 10,
  "direction": "ASC",
  "sortField": "createdAt"
}
```

Listing 11: Payload of an service-definition-query request.

3.10 PUBLISH arrowhead/serviceregistry/management/service-definition-create

Operation: service-definition-create
Input: ServiceDefinitionListRequest
Output: ServiceDefinitionListResponse

Creates service definitions.

```
{
  "serviceDefinitionNames": [
    "temperatureReading"
  ]
}
```

Listing 12: Payload of an service-definition-create request.

3.11 PUBLISH arrowhead/serviceregistry/management/service-definition-remove

Operation: service-definition-remove
Input: Array<String>

Removes service definitions by name, with their service instances.

```
[
  "TemperatureSensor"
]
```

Listing 13: Payload of an service-definition-remove request.

3.12 PUBLISH arrowhead/serviceregistry/management/service-query

Operation: service-query
Input: ServiceInstanceQueryRequest
Output: ServiceInstanceListResponse

Returns a page of service instances matching the filter, regardless of the discovery policy.
Optional entries of `params`: `verbose`.

```
{
  "pagination": {
    "page": 0,
    "size": 10,
    "direction": "ASC",
    "sortField": "createdAt"
  },
  "instanceIds": [
    "TemperatureSensor|temperatureReading|1.0.0"
  ],
  "providerNames": [
    "TemperatureSensor"
  ],
  "serviceDefinitionNames": [
    "temperatureReading"
  ],
  "versions": [
    "1.0.0"
  ],
  "alivesAt": "2026-10-22T12:00:00Z",
  "metadataRequirementsList": [
    {
      "unit": "celsius"
    }
  ],
  "addressTypes": [
    "IPV4"
  ],
  "interfaceTemplateNames": [
    "string"
  ],
  "interfacePropertyRequirementsList": [
    {
      "unit": "celsius"
    }
  ],
  "policies": [
    "string"
  ]
}
```

Listing 14: Payload of an service-query request.

3.13 PUBLISH arrowhead/serviceregistry/management/service-create

Operation: service-create
Input: ServiceInstanceCreateListRequest
Output: ServiceInstanceListResponse

Creates service instances on behalf of providers. The reservation of core and support service definitions applies.

```
{
  "instances": [
    {
      "systemName": "TemperatureSensor",
      "serviceDefinitionName": "temperatureReading",
      "version": "1.0.0",
      "expiresAt": "2026-10-22T12:00:00Z",
      "metadata": {
        "unit": "celsius"
      },
      "interfaces": [
        {
          "templateName": "generic_https",
          "protocol": "https",
          "policy": "CERT_AUTH",
          "properties": {
            "accessAddresses": [
              "192.168.0.10"
            ],
            "accessPort": 8080,
            "basePath": "/temperature",
            "operations": {
              "read": {
                "method": "GET",
                "path": "/read"
              }
            }
          }
        }
      ]
    }
  ]
}
```

Listing 15: Payload of an service-create request.

3.14 PUBLISH arrowhead/serviceregistry/management/service-update

Operation: service-update
Input: ServiceInstanceUpdateListRequest
Output: ServiceInstanceListResponse

Updates the expiry, metadata and interfaces of service instances.

```
{
  "instances": [
    {
      "instanceId": "TemperatureSensor|temperatureReading|1.0.0",
      "expiresAt": "2026-10-22T12:00:00Z",
      "metadata": {
        "unit": "celsius"
      },
      "interfaces": [
        {
          "templateName": "generic_https",
          "protocol": "https",
```

```
          "policy": "CERT_AUTH",
          "properties": {
            "accessAddresses": [
              "192.168.0.10"
            ],
            "accessPort": 8080,
            "basePath": "/temperature",
            "operations": {
              "read": {
                "method": "GET",
                "path": "/read"
              }
            }
          }
        }
      ]
    }
  ]
}
```

Listing 16: Payload of an service-update request.

3.15 PUBLISH arrowhead/serviceregistry/management/service-remove

Operation: service-remove
Input: Array<String>

Removes service instances by identifier.

```
[
  "string"
]
```

Listing 17: Payload of an service-remove request.

3.16 PUBLISH arrowhead/serviceregistry/management/interface-template-query

Operation: interface-template-query
Input: ServiceInterfaceTemplateQueryRequest
Output: ServiceInterfaceTemplateListResponse

Returns a page of interface templates.

```
{
  "pagination": {
    "page": 0,
    "size": 10,
    "direction": "ASC",
    "sortField": "createdAt"
  },
  "templateNames": [
    "generic_https"
  ],
  "protocols": [
    "https"
  ]
}
```

Listing 18: Payload of an interface-template-query request.

3.17 PUBLISH arrowhead/serviceregistry/management/interface-template-create

Operation: interface-template-create
Input: ServiceInterfaceTemplateListRequest
Output: ServiceInterfaceTemplateListResponse

Creates interface templates with their property rules.

```
{
  "interfaceTemplates": [
    {
      "name": "generic_https",
      "protocol": "https",
      "propertyRequirements": [
        {
          "name": "accessPort",
          "mandatory": true,
          "validator": "PORT",
          "validatorParams": [
            "string"
          ]
        }
      ]
    }
  ]
}
```

Listing 19: Payload of an interface-template-create request.

3.18 PUBLISH arrowhead/serviceregistry/management/interface-template-remove

Operation: interface-template-remove
Input: Array<String>

Removes interface templates by name.

```
[
  "TemperatureSensor"
]
```

Listing 20: Payload of an interface-template-remove request.

4 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Each *struct* denotes a JSON Object with the named fields; fields that are not needed may be omitted.

4.1 struct MqttRequestTemplate

The request envelope.

Object Field	Value Type	Description
"traceId"	String	Identifier chosen by the requester, returned in the response.
"authentication"	String	Identity of the requester (see Section 3).
"responseTopic"	String	Topic on which the response is published.
"qosRequirement"	Number	MQTT QoS of the response (0, 1 or 2).
"params"	Object<String>	Optional parameters (the query parameters of the HTTP binding).
"payload"	Object	The operation input.

4.2 struct MqttResponseTemplate

The response envelope.

Object Field	Value Type	Description
"status"	Number	Status code.
"traceId"	String	The trace identifier of the request.
"receiver"	String	Name of the requester.
"payload"	Object	The operation output, or an ErrorMessage.

4.3 struct ErrorMessage

Object Field	Value Type	Description
"errorMessage"	String	
"errorCode"	Number	
"exceptionType"	ExceptionType	
"origin"	String	

4.4 struct Address

A network address with its type.

Object Field	Value Type	Description
"type"	String	Address type: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`.
"address"	String	One network address (host name, IPv4, IPv6 or MAC address).

4.5 struct DeviceListRequest

Devices to create or update.

Object Field	Value Type	Description
"devices"	Array<DeviceRequest>	Devices to create or update.

4.6 struct DeviceListResponse

List of devices.

Object Field	Value Type	Description
"entries"	Array<DeviceResponse>	Matching records of the current page.
"count"	Number	Total number of matching records (all pages).

4.7 struct DeviceQueryRequest

Paged filter for devices.

Object Field	Value Type	Description
"pagination"	Page	Paging and sorting of the result.
"deviceNames"	Array<String>	Device names to match.
"addresses"	Array<String>	Network addresses of the entity.
"addressType"	String	Address type filter: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`.
"metadataRequirement List"	Array<MetadataRequirement>	Metadata requirements; each map is one alternative.

4.8 struct DeviceRequest

A device and its addresses.

Object Field	Value Type	Description
"name"	String	Device name, PascalCase.
"metadata"	Object	Free-form metadata (JSON object).
"addresses"	Array<String>	Network addresses of the entity.

4.9 struct DeviceResponse

A registered device.

Object Field	Value Type	Description
"name"	String	Device name.
"metadata"	Object	Free-form metadata (JSON object).
"addresses"	Array<Address>	Network addresses with their types.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

4.10 struct MetadataRequirement

A JSON object of metadata requirements: each key is a metadata path (dot notation), each value the required value or an operator object such as `{"op": "LESS_THAN", "value": 10}`.

4.11 struct Page

Paging and sorting; all fields optional.

Object Field	Value Type	Description
"page"	Number	Zero-based page number.
"size"	Number	Number of records per page (at most `max.page.size`, default 1000).
"direction"	String	Sort direction, `ASC` or `DESC`.
"sortField"	String	Field to sort by.

4.12 struct ServiceDefinitionListRequest

Service definitions to create.

Object Field	Value Type	Description
"serviceDefinition Names"	Array<String>	Service definition names.

4.13 struct ServiceDefinitionListResponse

List of service definitions.

Object Field	Value Type	Description
"entries"	Array<ServiceDefinitionResponse>	Matching records of the current page.
"count"	Number	Total number of matching records (all pages).

4.14 struct ServiceDefinitionResponse

A service definition.

Object Field	Value Type	Description
"name"	String	Service definition name.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

4.15 struct ServiceInstanceCreateListRequest

Service instances to create on behalf of providers.

Object Field	Value Type	Description
"instances"	Array<ServiceInstanceRequest>	Service instances to create or update.

4.16 struct ServiceInstanceInterfaceRequest

One interface of a service instance.

Object Field	Value Type	Description
"templateName"	String	Name of the interface template, snake_case (e.g. `generic_https`).
"protocol"	String	Protocol of the interface, e.g. `http`, `https`, `mqtt`, `mqtts`.
"policy"	String	Security policy of the interface (`NONE`, `CERT_AUTH`, token policies).
"properties"	Object	Interface properties defined by the template, e.g. `accessAddresses`, `accessPort`, `basePath`, `operations`.

4.17 struct ServiceInstanceInterfaceResponse

One interface of a service instance.

Object Field	Value Type	Description
"templateName"	String	Name of the interface template, snake_case (e.g. `generic_https`).
"protocol"	String	Protocol of the interface, e.g. `http`, `https`, `mqtt`, `mqtts`.
"policy"	String	Security policy of the interface: `NONE`, `CERT_AUTH` or one of the token policies.
"properties"	Object	Interface properties defined by the template, e.g. `accessAddresses`, `accessPort`, `basePath`, `operations`.

4.18 struct ServiceInstanceListResponse

List of service instances.

Object Field	Value Type	Description
"entries"	Array<ServiceInstanceResponse>	Matching records of the current page.
"count"	Number	Total number of matching records (all pages).

4.19 struct ServiceInstanceQueryRequest

Paged filter for service instances.

Object Field	Value Type	Description
"pagination"	Page	Paging and sorting of the result.
"instanceIds"	Array<String>	Instance identifiers to match.
"providerNames"	Array<String>	Provider system names to match.
"serviceDefinition Names"	Array<String>	Service definition names.
"versions"	Array<Version>	Service versions to match.
"alivesAt"	DateTime	Only instances that are still registered (not expired) at this UTC time are matched.
"metadataRequirements List"	Array<MetadataRequirement>	Requirements on the service instance metadata; each map is one alternative.
"addressTypes"	Array<String>	Address types the interfaces must offer (`HOSTNAME`, `IPV4`, `IPV6`, `MAC`).
"interfaceTemplate Names"	Array<String>	Accepted interface templates, e.g. `generic_https`.
"interfaceProperty RequirementsList"	Array<MetadataRequirement>	Requirements on the interface properties; each map is one alternative.
"policies"	Array<String>	Interface security policies to match, e.g. `CERT_AUTH`.

4.20 struct ServiceInstanceRequest

A service instance and its provider.

Object Field	Value Type	Description
"systemName"	String	Name of the provider system.
"serviceDefinitionName"	String	Service definition name, camelCase (e.g. `temperatureReading`).
"version"	Version	Version of the service instance (default `1.0.0`).
"expiresAt"	DateTime	UTC expiry time; no value means no expiry.
"metadata"	Object	Free-form metadata (JSON object).
"interfaces"	Array<ServiceInstanceInterface Request>	Service interfaces offered by the instance.

4.21 struct ServiceInstanceResponse

A registered service instance.

Object Field	Value Type	Description
"instanceId"	String	Service instance identifier `<System>|<serviceDefinition>|<version>`, or policy instance identifier.
"provider"	SystemResponse	The provider system.
"serviceDefinition"	ServiceDefinitionResponse	The service definition.
"version"	Version	Version of the service instance.
"expiresAt"	DateTime	UTC expiry time; no value means no expiry.
"metadata"	Object	Free-form metadata (JSON object).
"interfaces"	Array<ServiceInstanceInterface Response>	Service interfaces offered by the instance.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

4.22 struct ServiceInstanceUpdateListRequest

Service instances to update.

Object Field	Value Type	Description
"instances"	Array<ServiceInstanceUpdate Request>	Service instances to create or update.

4.23 struct ServiceInstanceUpdateRequest

Changes to one service instance.

Object Field	Value Type	Description
"instanceId"	String	Instance to update.
"expiresAt"	DateTime	UTC expiry time; no value means no expiry.
"metadata"	Object	Free-form metadata (JSON object).
"interfaces"	Array<ServiceInstanceInterface Request>	Service interfaces offered by the instance.

4.24 struct ServiceInterfaceTemplateListRequest

Interface templates to create.

Object Field	Value Type	Description
"interfaceTemplates"	Array<ServiceInterfaceTemplate Request>	Interface templates to create.

4.25 struct ServiceInterfaceTemplateListResponse

List of interface templates.

Object Field	Value Type	Description
"entries"	Array<ServiceInterfaceTemplate Response>	Matching records of the current page.
"count"	Number	Total number of matching records (all pages).

4.26 struct ServiceInterfaceTemplateProperty

Rule for one interface property.

Object Field	Value Type	Description
"name"	String	Name of the interface property.
"mandatory"	Boolean	Whether the property must be present in every interface of this template.
"validator"	String	Validator of the value, e.g. `PORT`, `NOT_EMPTY_ADDRESS_LIST`, `HTTP_OPERATIONS`.
"validatorParams"	Array<String>	Parameters of the validator.

4.27 struct ServiceInterfaceTemplateQueryRequest

Paged filter for interface templates.

Object Field	Value Type	Description
"pagination"	Page	Paging and sorting of the result.
"templateNames"	Array<String>	Interface template names to match.
"protocols"	Array<String>	Protocols to match.

4.28 struct ServiceInterfaceTemplateRequest

An interface template.

Object Field	Value Type	Description
"name"	String	Template name, snake_case.
"protocol"	String	Protocol of interfaces of this template.
"propertyRequirements"	Array<ServiceInterfaceTemplateProperty>	Property rules of the template.

4.29 struct ServiceInterfaceTemplateResponse

A stored interface template.

Object Field	Value Type	Description
"name"	String	Template name.
"protocol"	String	Protocol of the interface, e.g. `http`, `https`, `mqtt`, `mqtts`.
"propertyRequirements"	Array<ServiceInterfaceTemplateProperty>	Property rules of the template.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

4.30 struct SystemListRequest

Systems to create or update.

Object Field	Value Type	Description
"systems"	Array<SystemRequest>	Systems to create or update.

4.31 struct SystemListResponse

List of systems.

Object Field	Value Type	Description
"entries"	Array<SystemResponse>	Matching records of the current page.
"count"	Number	Total number of matching records (all pages).

4.32 struct SystemQueryRequest

Paged filter for systems.

Object Field	Value Type	Description
"pagination"	Page	Paging and sorting of the result.
"systemNames"	Array<String>	System names to match.
"addresses"	Array<String>	Network addresses of the entity.
"addressType"	String	Address type filter: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`.
"metadataRequirementList"	Array<MetadataRequirement>	Metadata requirements; each map is one alternative.
"versions"	Array<Version>	System versions to match.
"deviceNames"	Array<String>	Device names to match.

4.33 struct SystemRequest

A system on behalf of which an operator registers.

Object Field	Value Type	Description
"name"	String	System name, PascalCase.
"metadata"	Object	Free-form metadata (JSON object).
"version"	Version	Version of the system.
"addresses"	Array<String>	Network addresses of the entity.
"deviceName"	String	Name of the device hosting the system.

4.34 struct SystemResponse

A registered system.

Object Field	Value Type	Description
"name"	String	System name.
"metadata"	Object	Free-form metadata (JSON object).
"version"	Version	Version of the system.
"addresses"	Array<Address>	Network addresses with their types.
"device"	DeviceResponse	The device hosting the system, if registered.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

4.35 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available.

Type	Description
Array	A JSON Array [4]; Array<A> has elements of type A.
Boolean	JSON Boolean, `true` or `false`.
DateTime	JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`.
ExceptionType	One of `ARROWHEAD`, `INVALID_PARAMETER`, `AUTH`, `FORBIDDEN`, `DATA_NOT_FOUND`, `TIMEOUT`, `LOCKED`, `INTERNAL_SERVER_ERROR`, `EXTERNAL_SERVER_ERROR` (JSON String).
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.
Version	JSON String, semantic version `major.minor.patch`.

5 Go and Rust implementation variants

5.1 Go implementation

Primary system folder: `Eclipse-Arrowhead-go/ah5-core-go/serviceregistry`.

The system owns its Go module, launcher and build configuration. Use the per-system `Makefile` and `go.mod`; the containing Go tree is the build source, not a historical Arrowhead copy. Ordinary native roles use independent identities and JSON snapshot state rather than the Java database. Do not assume Java SQL migration, clustering or identical transaction/retry semantics. The registry has its own configuration and optional native AMQP monitor; other roles must be checked against their own registrations.

Listening port. Default 8443 (server.port in application.properties). Override with server.port, SERVER_PORT=N or -server.port N; the host is server.address. The native registry API, outsourced authentication, JSON persistence, optional AMQP telemetry and differences from Java are specified in `Eclipse-Arrowhead-go/ah5-core-go/serviceregistry/UPSTREAM-README.md`. It has no MQTT application API according to that profile. Native design/route evidence: `Eclipse-Arrowhead-go/ah5-core-go/serviceregistry/documentation`.

For this Java interface, select only a native transport and operation explicitly implemented and advertised by that port. A matching method/path does not establish DTO, error, authorization, retry or persistence equivalence. Java MQTT/AMQP bindings are not automatically provided by a Go or Rust role. Differential contract and security checks remain necessary before substitution.

5.2 Rust implementation

Primary system folder: `Eclipse-Arrowhead-rust/ah5-core-rust/serviceregistry`.

The per-system entry point is built by the `arrowhead-core-rust` crate. Build from the primary Rust workspace with the applicable package/binary selection. Documentation links can resolve into the shared core/support catalogue. Independent process identity and state remain required.

The ordinary native profile uses Tokio/Axum and rustls with peer-identity checks and independent locked JSON state. Network effects are not distributed database transactions. The precise trust, CRL reload and persistence guarantees are those of the native profile, not the Java implementation.

Native design/route evidence: `Eclipse-Arrowhead-rust/ah5-core-rust/documentation/ServiceRegistry`.

For this Java interface, select only a native transport and operation explicitly implemented and advertised by that port. A matching method/path does not establish DTO, error, authorization, retry or persistence equivalence. Java MQTT/AMQP bindings are not automatically provided by a Go or Rust role. Differential contract and security checks remain necessary before substitution.

6 References

[1] A. Banks and R. Gupta, “Mqtt version 3.1.1,” OASIS, Standard, 2014.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “serviceRegistryManagement – Service Description (ServiceRegistry),” Eclipse Arrowhead, SD 5.3.0, 2026, serviceRegistry-v5.3.0-serviceRegistryManagement-SD.pdf.

[6] ——, “serviceRegistryManagement HTTP/TLS/JSON – Interface Design Description (ServiceRegistry),” Eclipse Arrowhead, IDD 5.3.0, 2026, serviceRegistry-v5.3.0-serviceRegistryManagement-HTTP-IDD.pdf.

7 Revision History

7.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

7.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	